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Paleomagnetism of Alkaline Complexes and Remagnetization in the Kapuskasing Structural Zone, Ontario, Canada

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The Shenango and Nemegosenda alkaline complexes of the southern Kapuskasing Structural Zone (KSZ), an uplifted crustal block in northern Ontario, preserve primary thermoremanent magnetizations, with mean directions D=112°, I=–46° (k=13, α_{95}=13°, N=9 sites) and D=304°, I=+51° (k=31, α_{95}=10°, N=7 sites) for Shenango and D=124°, I=–57° (k=13, α_{95}=20°, N=4 sites) and D=312°, I=+54° (k=17, α_{95}=13°, N=7 sites) for Nemegosenda. Their average paleomagnetic pole is 45°N, 189°E (dp=6°, dm=8°), falling on the Keweenawan track of the North American apparent polar wander path (APWP) between 1098-1107 Ma reverse-polarity (R) and 1087-1098 Ma normal-polarity (N) poles of Lake Superior lavas and dikes. Unlike the Keweenawan paleopoles, which have a reversal asymmetry of ≈30°, the Shenango and Nemegosenda R and N poles are almost 180° reversed, which was found also by Lewchuk and Symons (1990b). This symmetry of R and N poles casts doubt on the previously accepted predominance of asymmetric reversals of the Earth's field in Keweenawan time. The Archean Shawmere anorthosite is overprinted by a B magnetization with mean direction D=97°, I=–68° (k=12, α_{95}= 10°, N=18 sites). The Shawmere B paleopole at 38°N, 226°E (dp=13°, dm=16°) falls between the carbonatite poles and ≈1850 Ma cratonic poles, and the B magnetization seems to be a composite of middle and late Proterozoic overprints. Sites in the western part of the Shawmere body close to the "carbonatite line" of faults linking the various alkaline complexes are severely altered, with an abundance of secondary magnetite and hematite. Virtual geomagnetic poles for these sites fall close to the Keweenawan track and have an average pole of 189°E, 53°N (dp=15°, dm=20°), suggesting that the B NRM at these sites is an ≈1100 Ma thermochemical overprint caused by hydrothermal fluids at the time of carbonatite intrusion. More easterly sites are less altered. Their VGPs resemble ≈1850 Ma cratonic poles and average to 235°E, 30°N (dp=15°, dm=17°). These B natural remanent magnetizations are probably thermochemical remanent magnetizations impressed by middle Proterozoic hydrothermal activity associated with intrusion of early alkaline bodies, fault movement on the Ivanhoe Lake fault zone, or late-stage uplift of the KSZ.

INTRODUCTION

The Kapuskasing Structural Zone (KSZ), a 400-km-long NNE-trending belt of high-grade gneissic rocks cutting across the E-W grain of greenstone belts in northern Ontario, is believed to be an uplifted cross section through the upper and middle Archean crust [*Percival and Card*, 1983; *Boland et al.*, 1988; *Percival et al.*, 1989; *Geis et al.*, 1990]. In previous paleomagnetic work [*Costanzo-Alvarez and Dunlop*, 1988] on amphibolite-grade tonalitic gneisses and the granulite-grade Shawmere anorthosite of the southern KSZ, we isolated two natural remanent magnetization (NRM) vectors. Both rock types had A magnetizations, with NE declinations and intermediate downward inclinations, but the tonalites and anorthosites required tilt corrections to move their paleomagnetic poles on to the late Archean apparent polar wander path (APWP) for Laurentia. The A NRM was thus acquired before regional tilting, probably about 2550 Ma ago.

The B magnetization, with easterly declination and steep to intermediate up inclination, was confined to the Shawmere anorthosite and had generally lower blocking temperatures than did A. We proposed that B was a thermochemical overprint related to intrusion of 1100-1050 Ma alkaline plugs in the KSZ and coeval with Keweenawan rifting and volcanism just to the west in Lake Superior. Our evidence was (1) the B paleopole falls close to the Keweenawan (1100-1050 Ma) track of the APWP, (2) the B component was more prominent at anorthosite sites close to the Nemegosenda carbonatite (age: 1107 Ma) [*Heaman and Machado*, 1992], and (3) *Lopez-Martinez and York* [1990] and *Hanes et al.* [1992] have reported 1200-1000 Ma K/Ar and $^{40}Ar/^{39}Ar$ feldspar ages from the southern KSZ. *Symons and Vandall* [1990], on the other hand, have interpreted the B as dating from the time of uplift and cooling of the KSZ at ≈1880 Ma, based on the fact that the B paleopole also resembles paleopoles of this age from the Superior Province.

The present paleomagnetic study of three Keweenawan-age alkaline complexes (Shenango, Nemegosenda, and Lackner Lake), of the remagnetized contact rocks of a fourth complex (Borden) in the southern KSZ, and of additional sites in the Shawmere anorthosite (Figure 1) is intended to test these rival hypotheses about the origin of the B overprint. We also report mineralogical and magnetic properties of Shawmere samples which delineate the areal extent of carbonatite-related hydrothermal alteration.

GEOLOGICAL SETTING AND PALEOMAGNETIC SAMPLING

Shawmere Anorthosite

The 15 x 50 km main body of the Shawmere anorthosite contains a border zone of migmatitic garnetiferous amphibolite; a banded zone of anorthosite, gabbro, garnet-rich, and ultramafic

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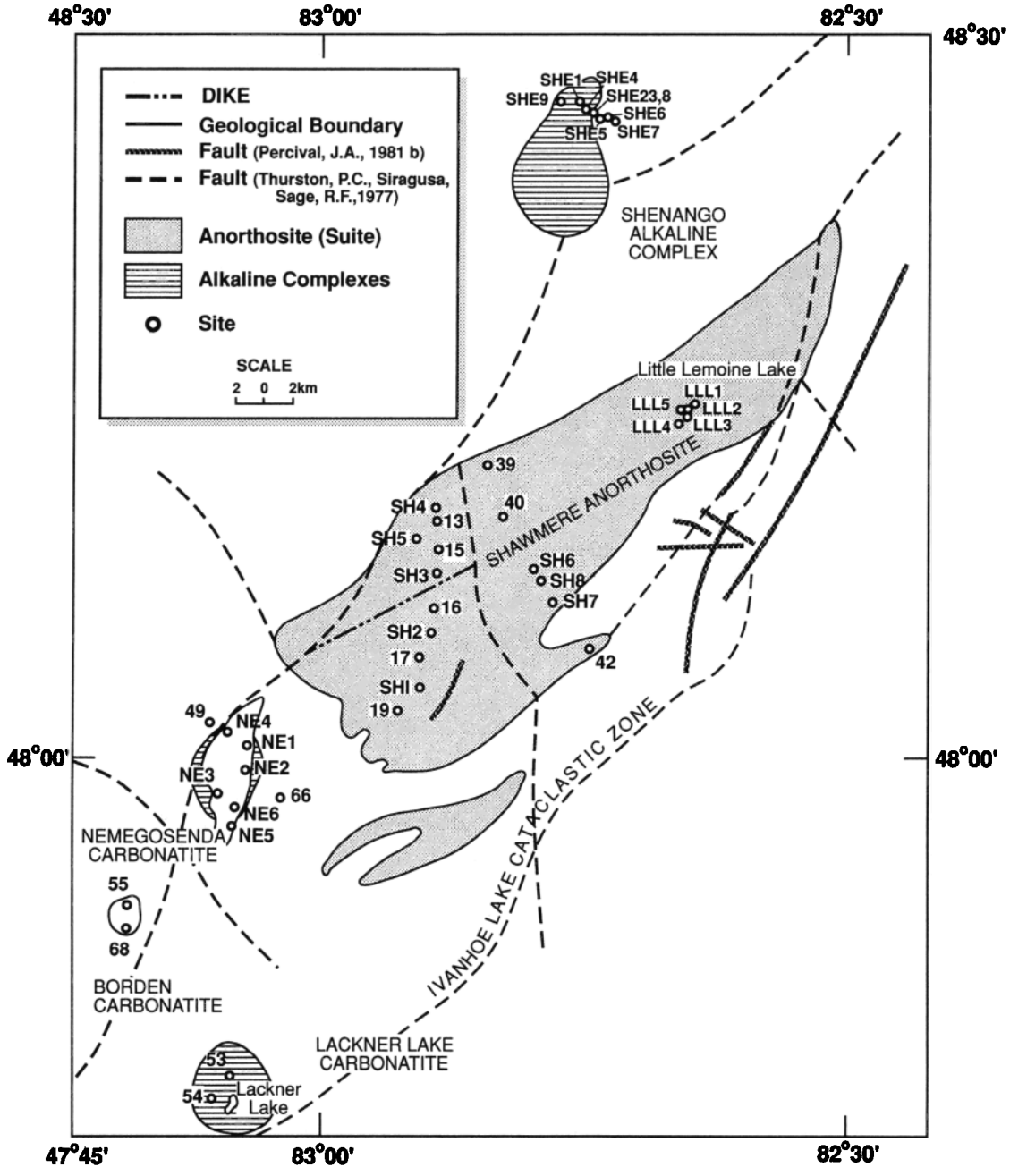


Fig. 1. Sketch map of the alkaline complexes and the Shawmere anorthosite in the Chapleau block showing sampling sites in these units. New sampling sites in the anorthosite are indicated along with the previous sites for the Shawmere River traverse from *Costanzo-Alvarez and Dunlop* [1988]. The faults mapped by *Thurston et al.* [1977] and *Percival* [1981] form a network of zones of weakness that could act as conduits for hydrothermal circulation in the Shawmere anorthosite. The "carbonatite line" discussed in the text corresponds to the fault linking the Borden, Nemegosenda, and Shenango complexes and cutting the SW corner of the Shawmere anorthosite.

rocks; an anorthosite zone with minor gabbro; and a zone of coarse-grained gabbroic anorthosite [*Percival and Card*, 1983]. The Shawmere anorthosite was probably intruded into mafic gneisses and paragneisses prior to 2765 Ma [*Percival and Krogh*, 1983], based on a U/Pb zircon date from a foliated garnetiferous tonalite within the anorthosite.

Oriented block samples for paleomagnetic study were taken from new sites in the north along the shore of Little Lemoine Lake (sites LLL1-LLL5), along a lumber road ≈20 km SW of Little Lemoine Lake (SH6-SH8, SH39-SH42), and in the south along the Shawmere River traverse of *Costanzo-Alvarez and Dunlop* [1988] (new sites SH1-SH5).

Alkaline Complexes in the Southern KSZ

Alkaline complexes and associated carbonatites in the Superior Province form one of the largest and oldest provinces of alkaline magmatism on Earth. The alkaline complexes are mainly unmetamorphosed circular bodies emplaced around 600, 1100, or 1800 Ma ago (whole rock (WR) K/Ar ages [*Gittins et al.*, 1967]). Associated carbonatites are relatively cold intrusions (680°-450°C [*Heinrich*, 1966]) which were rapidly emplaced (low contamination with country rock trace elements [*Bell et al.*, 1982, 1987]). The magmas probably originated in the subcontinental upper mantle (≥70 km).

In the Chapleau block of the southern KSZ, there are four alkaline intrusions: (1) the Shenango complex (1045 Ma by Rb/Sr WR [*Bell et al.*, 1982]; 1076 and 1082 Ma by K/Ar on biotite (unpublished data of K. Bell et al. reported by *Thurston et al.* [1977]); (2) the Nemegosenda carbonatite (1107 Ma by U/Pb on zircon [*Heaman and Machado*, 1992]; 1015 Ma by Rb/Sr WR [*Bell et al.*, 1982]; 1036 and 988 Ma by K/Ar on nepheline [*Gittins et al.*, 1967]); (3) the Lackner Lake carbonatite (1105 Ma by U/Pb on zircon [*Heaman and Machado*, 1992]; 1090-1095 Ma by K/Ar and Rb/Sr WR [*Gittins et al.*, 1967; *Bell et al.*, 1982]); and (4) the Borden carbonatite (1810 Ma by Rb/Sr WR and 1870-1890 Ma by U/Pb and Pb/Pb [*Bell et al.*, 1987]).

A line connecting these four bodies roughly parallels the Ivanhoe Lake Fault Zone (ILFZ), the surface expression of the basal decollement along which the KSZ was uplifted [*Geis et al.*, 1990]. *Bell et al.* [1982, 1987] remarked that recurrent alkaline magmatism in the Superior Province seems to be controlled by large-scale structural features such as the KSZ. The role of zones of weakness like the KSZ in intraplate alkaline magmatism has also been well documented elsewhere [*Sykes*, 1978]. Such zones of weakness could also act as channels for hydrothermal fluids accompanying the emplacement of the alkaline bodies.

The ≈1100-1050 Ma ages of the Shenango, Nemegosenda, and Lackner Lake bodies suggest a causal relationship with Keweenawan events in the Lake Superior area immediately to the southwest. *Burke and Dewey* [1973] proposed that the KSZ was a failed third arm of a Keewenawan triple junction. In fact, the KSZ is a much older structure that has been repeatedly reactivated [*Hanes et al.*, 1992]. In response to the tectonic forces that caused Keweenawan rifting, the KSZ became the site of incipient rifting and alkaline intrusion.

The Shenango complex is an elliptical body intruding mafic paragneisses and foliated dioritic rocks. The complex comprises a monzonite core with arcuate mafic hornfels units and a peripheral unit of syenodiorite and diorite on the east side. These units seem to be comagmatic and grade into one other. Unlike other alkaline complexes in the Chapleau block, the Shenango lacks the characteristic units of a carbonatite-alkalic suite and there is no fenitization of the surrounding country rocks.

Ours is the first paleomagnetic study of the Shenango body. Samples were taken at eight sites in the northern end of the intrusion, along the railway line and the shore of Shiners Lake. This part of the Shenango intrusion consists of massive fresh diorite with accessory magnetite. Some sites were also taken in the mafic gneiss country rock.

The Nemegosenda carbonatite consists of a nucleus of alkaline syenite and carbonatite, surrounded by brecciated syenitic rocks [*Parsons*, 1961; *Thurston et al.*, 1977]. Brecciation, alteration, and recrystallization increase outward to a metasomatic aureole. The Archean gneiss in the aureole shows two fenitization zones. The inner zone is characterized by dehydration and reduction with abundant magnetite, and the foliated outer zone is extensively hydrated and oxidized and rich in hematite. Minor late mafic, porphyry, and carbonatite dikes and carbonatite veins occur in and near the complex.

Symons and Garber [1974] carried out a paleomagnetic study of 17 sites in different petrological phases in the complex and its aureole. They reported a normal (N) polarity remanence (D=305°, I=+46°), whose corresponding paleopole (178°W, 43°N) falls among other N poles on the 1100-1050 Ma Logan loop or Keweenawan track of the APWP [*Halls and Pesonen*, 1982]. We augmented Symons and Garber's study with eight new sites: NE-1, NE-2 and NE-6 were in alkalic syenite, site 3 was in carbonatite at the nucleus of the complex, sites 4, 49, and 66 were in the contact aureole, and site 5 was in syenitic breccia.

The Lackner Lake carbonatite, located adjacent to the ILFZ, has a diameter of 5.5 km and consists mainly of coarse-grained nepheline syenite that has fenitized the Archean granulite gneisses it intrudes [*Hodder*, 1961; *Parsons*, 1961; *Thurston et al.*, 1977]. Carbonatite and apatite-magnetite veins, as well as occasional dikes known from aeromagnetic maps, cut the syenite. Apart from minor faults on the south and east sides, there is no evidence of postintrusion tectonism.

Symons [1989*a*] reported a mean NRM direction D=305°, I=+64° for the Lackner Lake carbonatite, with a corresponding paleopole at 157°W, 54°N. As with Nemegosenda, there were no reverse (R) polarity results. Symons' pole falls on the Keweenawan track but in a position somewhat younger than the Nemegosenda pole. Because of poor exposure we took only two additional sites (53 and 54) in the Lackner Lake body.

The Borden carbonatite is marked by a prominent circular magnetic anomaly with a diameter of 1.6 km. There are no surface outcrops, but exploration trenches and drill holes suggest that the main lithologies are fenite, calcic carbonatite, and silico-carbonatites [*Bell et al.*, 1987]. We sampled two sites for paleomagnetic study in the tonalitic country rock overlying the intrusion.

EXPERIMENTAL METHODS

We collected an average of six samples per site. Samples were oriented by sun and Brunton compasses. Alternating field (AF) demagnetization in ≈20 steps to 100 mT and thermal demagnetization in ≈15 steps, usually to 600°C, were carried out in magnetic shields with standard Schonstedt demagnetizers. The NRM vectors of thermally cleaned specimens were measured with a Schonstedt spinner magnetometer (sensitivity ≈ 0.05 mA/m). A modified Digico spinner magnetometer with similar sensitivity was used to measure the rest of the specimens. NRM directions for 21 Nemegosenda and Shenango samples were measured at the Geological Survey of Finland, using a Schonstedt AF demagnetizer with tumbler and a μ-metal shielded coil-spinner magnetometer (sensitivity ≈ 0.2 mA/m) [*Mertanen et al.*, 1987].

NRM components were defined by linear trajectories on orthogonal vector projections [*Zijderveld*, 1967]. The trajectories were tested by principal component analysis [*Kirschvink*, 1980] and were accepted only if a segment including at least three consecutive points (not including the origin) had an angular uncertainty ≤10°. The characteristic (highest coercivity H_c or unblocking temperature T_{UB}) NRM for each specimen was revealed as a stable end point direction on stereographic projections. The number of stable samples was small at some sites (mainly in the carbonatite complexes) because of the dominance of soft multidomain remanence carriers.

Initial volume susceptibilities, which were used to calculate Koenigsberger Q-ratios (intensity of NRM/intensity of induced magnetization), were measured for all the anorthosite samples using a Bartington ac bridge with a peak field of ≈0.1 mT.

PALEOMAGNETIC RESULTS

The B magnetization of *Costanzo-Alvarez and Dunlop* [1988] was isolated by AF and thermal demagnetization in 14 new sites in the Shawmere anorthosite. An NRM with a similar although not identical direction was found in all four alkaline complexes. For convenience, we shall refer to this carbonatite NRM as a B magnetization also. In some cases, both normal-polarity B (B_N) and reverse-polarity B (B_R) were present in a single sample, but usually either B_N or B_R dominated the NRM after removal of a soft viscous overprint in the direction of the present Earth's field (PEF). Thermal decay curves indicated almost pure magnetite

with blocking temperatures near 580°C as the main carrier of the B remanence in the Shawmere anorthosite, with hematite as a minor carrier in some cases. B unblocking temperatures in the alkaline complexes spanned a wider range of values, revealing differences in their magnetic mineralogies and grain sizes compared to those of the anorthosite. Representative paleomagnetic results will be discussed separately for each unit.

Shawmere Anorthosite

The B remanence is the characteristic (highest H_c/T_{UB}) NRM of most anorthosite samples (18 sites, 60 samples). Figure 2 illustrates the AF and thermal demagnetization response of some individual specimens. The stable B_R remanence is carried by magnetic minerals with high coercivities and unblocking temperatures (fine-grained magnetite and hematite). Hematite is evident from the incomplete demagnetization of B_R at 580°C (Figures 2*b* and 2*d*). Minor PEF overprints are usually erased by AF cleaning to 6 mT.

Sample 42-5 (Figure 2*c*) has a bivectorial NRM. The low-coercivity remanence of this sample could be either B_N or a PEF overprint; it is completely erased by an AF of 30 mT. The underlying B_R remanence has a hard coercivity spectrum (calculated from the Zijderveld plot using the graphical method of *Dunlop* [1979]) that resembles spectra of single-component B_R samples. However, B_R accounts for <25% of the total NRM of sample 42-5.

Site mean directions for B_R are plotted in Figure 5*a* and listed in Table 1. Data from four previous sites [*Costanzo-Alvarez and Dunlop*, 1988] are included in the calculation of the final average (D=97°, I=−68°, k=12, α_{95}=9.5°, N=18 sites).

Shenango Alkaline Complex

Both B_N and B_R remanences were isolated in samples from the Shenango complex. In most cases, the NRM was a stable B_R remanence carried by high-H_c and high-T_{UB} magnetic minerals (fine-grained magnetite and hematite). B_N is typically a lower-H_c, low-T_{UB} remanence, probably carried by coarse-grained multi-domain (MD) magnetite (Figure 3*d*). SHE-611 (Figure 3*a*) is an example of a B_N remanence carried by both high- and low-H_c minerals. In sample SHE-832 (Figure 3*b*), removal of a PEF overprint reveals an underlying high-H_c (50-100 mT) B_R remanence. Figures 3*c* and 3*d* give examples of superimposed B_N and B_R magnetizations. Sample SHE-652 (Figure 3*c*) is unusual in that B_R has coercivities in the 4-50 mT range and is accompanied by a small but very hard B_N component that is barely affected by 80 mT cleaning. Sample SHE-222 (Figure 3*d*) is more typical. The B_N remanence is soft, while B_R has coercivities from 30 to >80 mT and blocking temperatures of 520°-575°C. In this sample, B_R is almost twice as strong as B_N and is carried about 2/3 by SD magnetite with unblocking temperatures just below 580°C and 1/3 by hematite.

Site mean B_R and B_N directions are plotted in Figure 5*b* and listed in Table 2. Final averages are D=112°, I=−46°, (k=13, α_{95}=13°, N=9 sites) for B_R and D=304°, I=+51°, (k=31, α_{95}=10°, N=7 sites) for B_N.

Nemegosenda Carbonatite

Figure 4 gives some examples of the AF demagnetization response of B_N and B_R remanences. In NE-611 (Figure 4*a*), B_N is the only remanence and demagnetizes over a wide range of coercivities. In carbonatite sample NE-342 (Figure 4*b*), after removal of a large PEF overprint B_R is found to have an AF decay curve with an initial plateau, an inflected shape, and generally high destructive fields typical of SD magnetite. In complete contrast, in fenitized gneiss sample NE-421 (Figure 4*c*), B_R is carried by MD magnetite, as evidenced by the exponential decay of NRM in AF cleaning.

Site mean B_N and B_R directions are plotted in Figure 5*c* and listed in Table 3. Final averages are D=124°, I=−57°, (k=13, α_{95}=20°, N=4 sites) for B_R and D=312°, I=+54°, (k=17, α_{95}=13°, N=7 sites) for B_N.

Borden and Lackner Lake Carbonatites

There are relatively few results for these two intrusions, and those for the Borden come from the overlying country rocks. The Borden data resemble B_N results for the other alkaline complexes (Figure 5 and Table 4). Hence the country rocks overlying the Borden body probably were remagnetized at the time the younger carbonatites were intruded. For the Lackner Lake carbonatite, in contrast with *Symons'* [1989*a*] exclusively N-polarity data, R-polarity remanences (B_R) were isolated in three samples from two of the sites sampled (Figure 5*d*). Although Lackner and Borden results are scanty, they support conclusions drawn from the Shenango and Nemegosenda data.

DISCUSSION

Paleopoles and Apparent Polar Wander Path

Paleopoles corresponding to B_N and B_R final means from the Shenango and Nemegosenda alkaline complexes fall on the North American apparent polar wander path (APWP) (Figure 6), midway between N- and R-polarity groups of paleopoles from Keweenawan units of the Lake Superior region, with U/Pb zircon ages of 1087-1098 Ma and 1098-1107 Ma, respectively (see Table 5). The Shenango and Nemegosenda NRMs are thus tightly bracketed in age. They are primary thermoremanent magnetizations (TRMs) acquired in initial cooling of the Shenango and Nemegosenda complexes around 1100 Ma or thermal overprints (partial TRMs) of adjacent country rocks. Notice that the B_N and B_R poles do not segregate into distinct N and R groups like the Keweenawan results but are mutually concordant.

Our Nemegosenda paleopoles are not significantly different from the N-polarity pole of *Symons and Garber* [1974]. Also plotted in Figure 6 are N-polarity poles for other ≈1100-Ma alkaline complexes of the KSZ, the Lackner Lake carbonatite [*Symons*, 1989*a*], the Firesand River carbonatite [*Symons*, 1989*b*], the Clay-Howells alkaline complex [*Lewchuk and Symons*, 1990*a*], and N and R poles for the 1108 Ma (U/Pb on zircon) Coldwell alkaline complex [*Lewchuk and Symons*, 1990*b*] and the Chipman Lake and Seabrook Lake carbonatite complexes [*Symons*, 1992] bordering Lake Superior just west of the KSZ. These also fall on the Keweenawan track but in positions slightly older or younger than the Shenango and Nemegosenda poles.

Because of the present controversy about whether the large difference in N and R pole positions for Keweenawan units represents genuine APW or is an artefact of asymmetric field reversal, it is not yet clear whether Figure 6 can be used to "fine tune" magnetization ages of the various alkaline intrusions in and near the KSZ. Although the nondipole model of *Nevanlinna and Pesonen* [1983] does permit a succession of reversals of differing degrees of asymmetry, the discovery of symmetric reversals in the Shenango and Nemegosenda complexes and in the Coldwell, Chipman, and Seabrook Lake complexes makes it unlikely that asymmetric reversals were the norm in Keweenawan time.

The B_R paleopole for the Shawmere anorthosite falls between 1098-1107 Ma Keweenawan R poles and a group of ≈ 1850 Ma old poles from the Superior Province (MD, SG, SN, and SR in

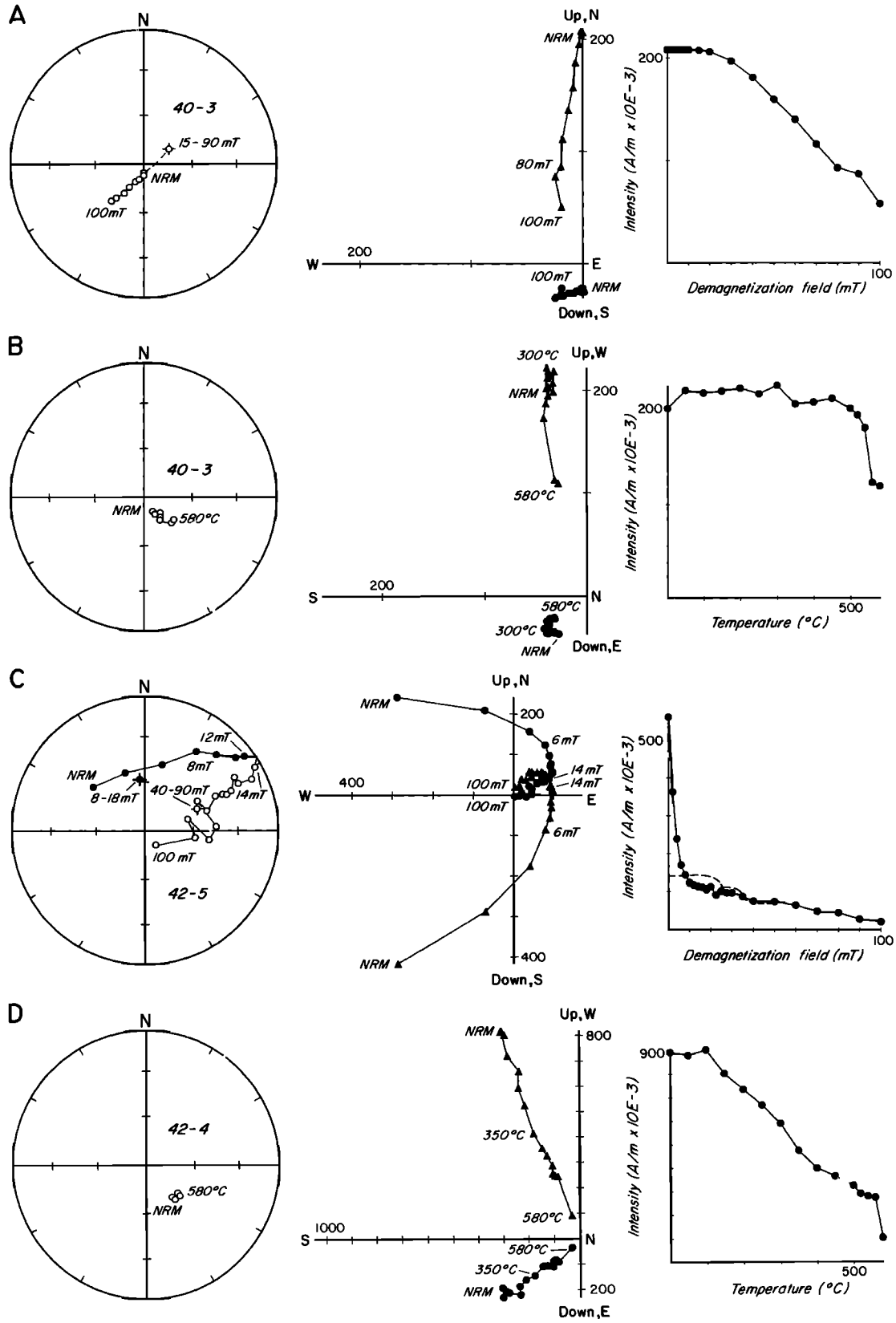


Fig. 2. Typical high unblocking temperatures and coercivities of the B_R natural remanent magnetization (NRM) in the Shawmere anorthosite samples. Equal-area and orthogonal vector projections of magnetization vectors in the course of alternating field and thermal demagnetization are shown for samples 40-3, 42-5, and 42-4. In the orthogonal vector projections, circles are horizontal plane projections and triangles are vertical plane projections. Intensity decay curves are shown for each sample. A superimposed dashed curve corresponding to the intensity decay of the underlying B_R NRM [*Dunlop*, 1979] is sketched for sample 42-5.

TABLE 1. B_R Site and Grand Mean Directions for the Shawmere Anorthosite

Site	Distance to Carbonitite Line, km	Declination, deg	Inclination, deg	N/N'	α_{95}, deg	Fisher precision, k
42	16	110	-62	3/5	19	14
LLL1	11.7	330	-67	3/5	11	52
LLL2	11.7	119	-80	4/5	17	17
LLL3	11.7	63	-66	4/5	31	5
LLL4	11.7	76	-73	3/5	16	25
LLL5	11.7	127	-77	4/8	14	24
SH7	11.7	67	-76	5/6	18	12
SH6	9.3	88	-34	5/8	10	39
19	7	131	-82	5/7	11	35
17	6	55	-67	5/5	12	28
SH2	6	71	-42	2/6		15
40	5.7	132	-80	2/5		37
SH3	4.3	57	-42	2/6		54
15	3.3	121	-56	3/6	14	36
39	2.7	132	-31	1/5		
13	2	139	-53	1/6		
SH4	1.7	163	-74	2/7		41
SH5	1.7	113	-64	4/5	16	18
Mean		97	-68	18	9.5	12

N' is the total number of samples per site; N is the number with stable behavior.

Figure 6). The B paleopole of *Symons and Vandall* [1990], which was based on both gneiss and anorthosite samples, falls south and east of our Shawmere pole and is quite similar to the MD and SG poles. A B-like pole from Kapuskasing dikes [*Bates and Halls*, 1991] falls well to the east of the Shawmere pole and north of the SN pole. In both studies, it was proposed that B magnetizations in the KSZ were acquired during uplift of the Chapleau and Groundhog River blocks between ≈2000 and ≈1900 Ma ago. Symons and Vandall assigned an age of ≈1880 Ma to their KSZ B overprint.

We are not convinced that this age assignment is correct for our Shawmere B results for several reasons. Unlike the Keweenawan event, the uplift of the KSZ is not precisely dated. It is not certain that this group (MD, SG, SN, and SR) is the appropriate set of cratonic reference poles. This set of poles, and middle Proterozoic poles from the craton in general, are poorly grouped and provide an uncertain reference pole. The Shawmere pole falls as close to Abitibi-Keweenawan poles as it does to the average of ≈1850 Ma Superior poles (Figure 6). In the following sections, we consider other evidence bearing on the timing of the B remagnetization in the Shawmere anorthosite.

Patterns in Shawmere B NRM Directions

Figures 5 and 6 show that there is a distinct difference between mean B NRM directions and paleopoles of the Shawmere anorthosite and those of the carbonatites and their remagnetized contact rocks. If the time calibration of the Keweenawan track is to be trusted, 30-40 Ma elapsed between remagnetization of the Shawmere and the time the alkaline complexes were intruded and acquired primary TRM.

It is possible to reconcile the Shawmere and carbonatite data without invoking a remagnetization event 30-40 Ma prior to carbonatite intrusion if there are actually two different B magnetizations in the Shawmere anorthosite, one about 1100 Ma in age with shallower inclination like the Shenango and Nemegosenda NRMs (Figures 5*b* and 5*c*) and the other dating from the middle Proterozoic with steeper inclination like *Symons and Vandall's* [1990] B magnetization. There is some suggestion in Figure 5*a* of streaking of site mean directions between shallow and steep inclinations. We note that Symons and Vandall concentrated their sampling along Highway 101 close to the ILFZ, while we sampled farther west toward the "carbonatite line" linking the Borden, Nemegosenda, and Shenango complexes (Figure 1). A major fault follows this line [*Thurston et al.*, 1977] and cuts the SW corner of the Shawmere.

Suppose that the steeper B NRM is connected to middle Proterozoic uplift on the ILFZ and/or to ≈1850 Ma hydrothermal activity in the KSZ recently dated by *Hanes et al.* [1992] using muscovite and biotite $^{40}Ar/^{39}Ar$ ages, whereas the shallower B NRM is linked to ≈1100 Ma alkaline magmatism and associated hydrothermal activity near the carbonatite line, detected by 1200-1000 Ma K/Ar and $^{40}Ar/^{39}Ar$ feldspar ages [*Lopez-Martinez and York*, 1990; *Hanes et al.*, 1992]. Then we would expect the following patterns.

1. Sites close to the carbonatite line should be strongly remagnetized with a shallow B NRM.
2. Sites close to the ILFZ should have strong steep B overprints.
3. Sites intermediate between the carbonatite line and the ILFZ should have less tendency to be altered and remagnetized.

There is some evidence for such trends. Intermediate sites SH-1 and SH-8 (7.7 and 10 km, respectively, from the carbonatite line) have only A magnetizations and have not been remagnetized at all. On average, the B inclination is shallower at sites close to the carbonatite line and steeper at sites closer to the ILFZ, although there is a large dispersion of inclinations. More convincing is the fact that VGPs for the four sites closest to the carbonatite line fall at or near 1100 Ma on the Keweenawan track with an average pole B_{R2} of 189°E, 53°N (dp=15°, dm=20°), while the remaining site means give much more easterly VGPs, some falling in the

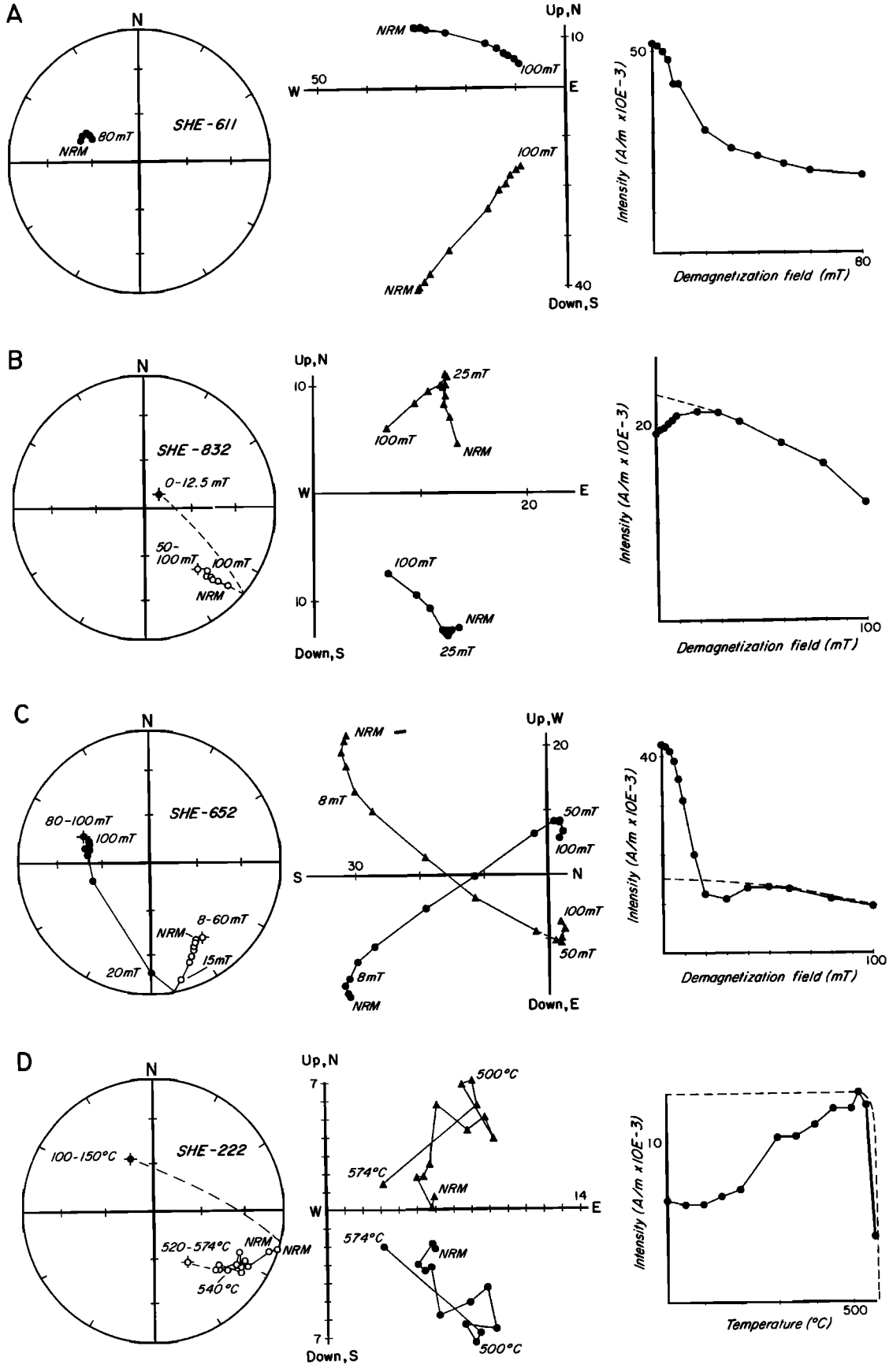


Fig. 3. Magnetic unblocking temperatures and coercivities of the B_N and B_R natural remanent magnetizations (NRMs) in the Shenango alkaline complex samples 611, 832, 652, and 222. Superimposed dashed curves correspond to the intensity decay of B_R or B_N underlying characteristic NRMs.

TABLE 2. B_R and B_N Site and Grand Mean Directions for the Shenango Alkaline Complex

Site	Declination, deg	Inclination, deg	N/N'	α_{95}, deg	Fisher precision, k
SHE-1	119	-59	2/6		3800
SHE-2	135	-38	3/6	12	49
SHE-3	77	-65	1/5		
SHE-4	114	-52	3/5	30	7
SHE-5	123	-42	1/5		
SHE-6	125	-29	2/8		6
SHE-7	108	-24	1/6		
SHE-8	121	-33	3/6	16	25
SHE-9	51	-51	2/7	23	18
Mean	112	-46	9	13	13
SHE-1	281	50	2/6		194
SHE-2	339	60	3/6	14	36
SHE-3	307	34	2/5		
SHE-4	315	48	3/5	15	29
SHE-5	298	55	4/5	15	154
SHE-6	285	45	6/8	7	75
SHE-7	313	57	2/6		
Mean	304	51	7	10	31

N' is the total number of samples per site; N is the number with stable behavior.

≈1850 Ma grouping with an average paleopole B_{R1} at 235°E, 30°N (dp=15°, dm=17°) (see Figures 6 and 7).

The two hypothesized B_R remanences have very similar directions, and since they are both likely thermochemical overprints, their spectra of coercivities and unblocking temperatures must overlap substantially. The separation of two B_R remanences is therefore somewhat arbitrary. B_R remanences at some sites could be hybrids of the two coexisting remanences which cannot be resolved by routine demagnetization procedures. Another possibility is that hematite, a by-product of hydrothermal oxidation of primary magnetite, may carry a chemical remanent magnetization (CRM) whose direction is biased away from the Keweenawan field direction by the field of an older steep B_R remanence [e.g., *Heider and Dunlop*, 1987]. Such a CRM is not a hybrid but a univectorial remanence with a direction intermediate between the two B_R remanences.

Anisotropy of magnetic susceptibility (AMS) indexes k_{max}/k_{min} for 94 anorthosite samples peak around 1.15 (Figure 8). Such a low value indicates nearly isotropic magnetite as the main magnetic mineral [*Hyodo*, 1989]. This in turn rules out the possibility of explaining the variations of B_R inclinations in the Shawmere anorthosite as a consequence of magnetic fabric. *Symons and Vandall* [1990] concluded that magnetic fabric had no systematic effect on the direction of their B remanences even in highly anisotropic gneiss samples. Minor contamination by a PEF viscous overprint would have the effect of shallowing the B_R vector. Survival of any PEF overprint in the typical range of Shawmere B_R blocking temperatures (just below 580°C) is most unlikely, however.

Patterns in Mineral Alteration and Magnetic Properties

Mineralogical and magnetic properties of Shawmere anorthosite samples also vary with distance from the "carbonatite line", which we take to mark the approximate surface intersection of the incipient rift in which the carbonatite plugs were emplaced. The series of faults that follow the carbonatite line (Figure 1) probably acted as conduits for fluids during hydrothermal alteration. Shawmere anorthosite thin sections were examined at magnifications up to 400X in transmitted light with a polarizing microscope. Alteration takes the form of veins and cracks that cut pervasively through the primary plagioclase crystals, being more prominent in westerly sites close to the carbonatite line.

In a sample from site 39, 2.65 km from the carbonatite line (Figure 9*a*), opaque minerals occur as a powder of fine grains associated with alteration veins. A sample from site SH-2, 6 km from the carbonatite line (Figure 9*b*), contains very fine grained opaque minerals growing within fresh plagioclase crystals. Most of the Little Lemoine Lake samples, 11.7 km from the carbonatite line, (Figures 9*c* and 9*d*) are quite fresh. The few opaque minerals observed appear as a barely recognizable very fine powder, filling inclusions and small veins within the plagioclase crystals (Figure 9*c*), or as large isolated hematite or hematite-coated magnetite grains (Figure 9d).

Changes in the bulk magnetic properties of altered rocks, for example, decreases in susceptibility and remanent intensities, accompanied by increases in coercivities, blocking temperatures, and Koenigsberger ratios, have proven a valuable tool in mineral exploration and other applications [e.g., *Geissman et al.*, 1980; *Ueno*, 1982; *Criss and Champion*, 1984; *Lapointe et al.*, 1986]. We therefore supplemented our thin section observations with magnetic properties data. Average Koenigsberger Q ratios (Figure 10) are a means of mapping the extent of hydrothermal alteration in the Shawmere anorthosite. A higher degree of alteration results in abundant fine magnetite with single-domain (SD) values of Q. These sites are mainly the westerly ones (Figure 10), with a maximum Q around 4 km from the carbonatite line. There are also two easterly sites within 4 km of the ILFZ with SD Q values of 2-3, lower than the Q values of 8-15 at the westerly sites. Intermediate sites ≥6 km from either the carbonatite line or the ILFZ tend to have Q values ≤1

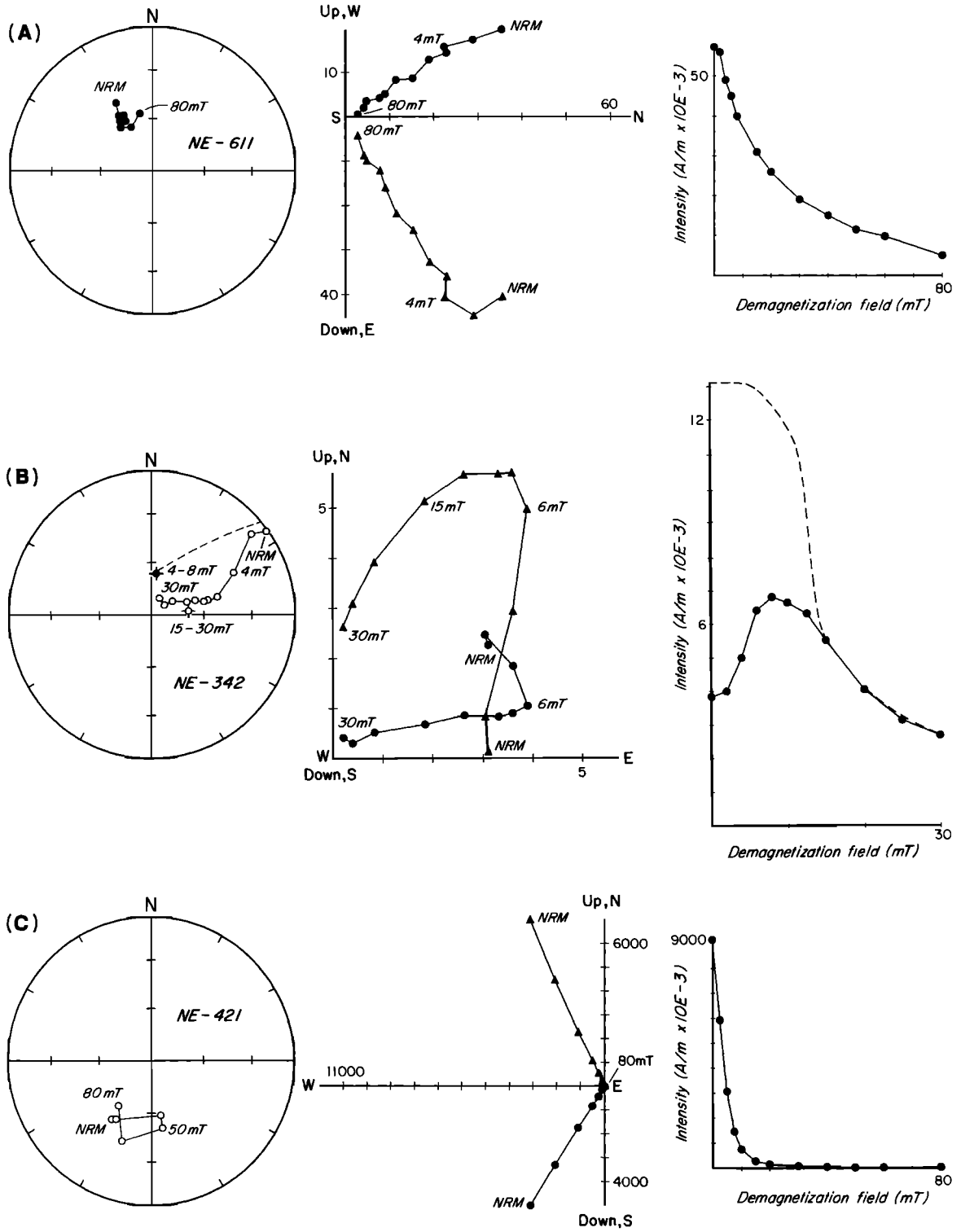


Fig. 4. B_N and B_R coercivities in Nemegosenda carbonatite samples NE-611, NE-342 and NE-421.

characteristic of unaltered or only surface-oxidized coarse magnetite in a MD state. The dispersion in Q values at a site also tends to be high at westerly sites, reflecting wider ranges of magnetite grain sizes in the altered anorthosites, and low for intermediate sites well away from mapped faults.

Variations of Koenigsberger ratios probably result both from oxidation of primary multidomain (MD) magnetites, which decreases their effective grain sizes, and from nucleation of secondary magnetite and hematite. These processes can be crudely simulated in a simple experiment that we carried out on fresh samples from sites LLL-2 and SH-1. Saturation isothermal remanent magnetization (SIRM) became harder to AF demagnetize after samples were heated for 1 hour at 700°C (Figure 11). Furthermore, the SIRM before heating decayed almost exponentially (linear on the log-linear plot in Figure 11) from 0 to 100 mT, suggesting dominantly large MD magnetite grains

TABLE 3. B_R and B_N Site and Grand Mean Directions for the Nemegosenda Carbonatite

Site	Declination, deg	Inclination, deg	N/N'	α_{95}, deg	Fisher precision, k
NE-1	135	-78	1/7		
NE-3	137	-57	1/6		
NE-4	119	-26	2/6		204
66	112	-65	1/4		
Mean	124	-57	4	20	13
NE-1	330	41	2/7		221
NE-2	271	66	1/5		
NE-3	266	46	2/6		25
NE-5	330	49	3/6	14	24
NE-6	320	45	3/5	11	50
49	340	54	3/7	16	25
66	310	59	3/4	20	16
Mean	312	54	7	13	17

N' is the total number of samples per site; N is the number with stable behavior.

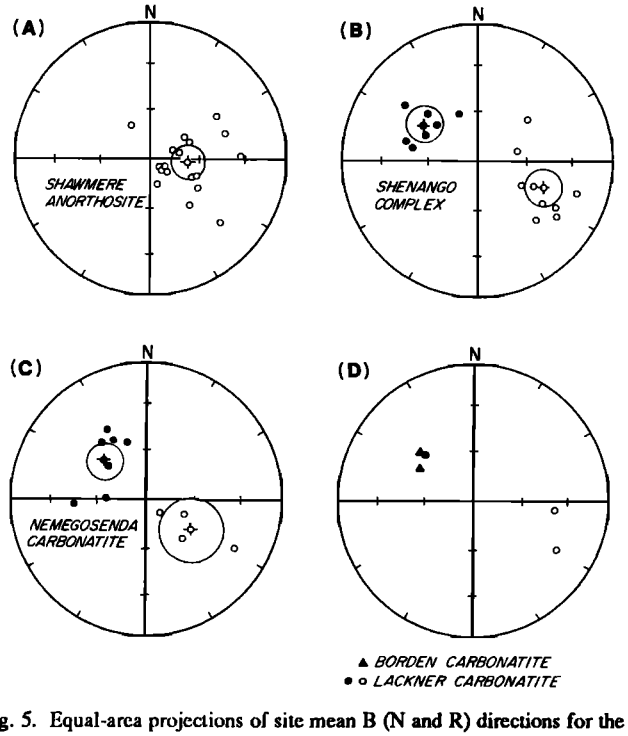


Fig. 5. Equal-area projections of site mean B (N and R) directions for the Shawmere anorthosite, the Shenango alkaline complex, and the Nemegosenda, Lackner Lake, and Borden carbonatites. Unit means (stars) and circles of 95% confidence are also shown for Shawmere, Shenango, and Nemegosenda.

[*Dunlop*, 1983; *Xu and Dunlop*, 1993], but after heating, the shift toward smaller magnetite grains sizes produced less exponential curves as well as higher average coercivities.

Inferred Magnetic and Geological History

The main boundary conditions on an interpretation of our results are the following.

1. The B NRMs of the Shenango and Nemegosenda alkaline intrusions are stable primary TRMs with both N and R polarities dating from Keweenawan time (≈1100 Ma). There are too few results from our Borden and Lackner Lake samplings to calculate reliable paleopoles, but their site mean VGPs are also on or near the Keweenawan track.

2. The average B NRM of the Shawmere anorthosite is a secondary overprint whose paleopole agrees with neither ≈1100 Ma Keweenawan poles nor ≈1850 Ma cratonic poles.

3. There is no direct evidence for "Keweenawan precursor" tectonic activity in the KSZ apart from rather imprecise 1200-1000 Ma $^{40}Ar/^{39}Ar$ feldspar ages, which may correspond to hydrothermal circulation associated with intrusion of the Shenango, Nemegosenda, and other ≈1100 Ma intrusions.

4. There is evidence for uplift of the KSZ in the middle Proterozoic, as a series of thrust sheets above a basal decollement which surfaces near the ILFZ [*Geis et al.*, 1990]. The timing is still under debate [*Percival et al.*, 1988; *Hanes et al.*, 1992], but a lower limit is ≈1880 Ma, since the Borden and Cargill alkaline complexes which are of this age are undisturbed [*Percival et al.*, 1988; *Symons and Vandall*, 1990]. $^{40}Ar/^{39}Ar$ mica ages of ≈1850 Ma [*Hanes et al.*, 1992] may date uplift fault motions or may be associated with hydrothermal circulation in the older (Borden/Cargill) episode of alkaline magmatism.

5. The fine-grained magnetite in Shawmere anorthosites, which can be expected to be the source of stable NRM, is associated with alteration (Figure 9).

6. All indicators of alteration (thin section observations and the magnitude and within-site dispersion of the Koenigsberger Q ratio) imply strongest alteration within ≈6 km of the "carbonatite line" which marks the site of a fault system that could have channelled hydrothermal fluids during 1100 Ma alkaline magmatism. Shawmere B NRMs from this westerly zone tend to have lower inclinations and ≈1100 Ma VGP positions. The same or a similar fault system could have channelled hydrothermal circulation in earlier times, but the alteration we see must date mainly from ≈1100 Ma because even the contact rocks overlying the 1870-1890 Ma Borden carbonatite bear an ≈1100 Ma, not an ≈1880 Ma, overprint (Figure 5*d*).

7. There is a more subdued alteration peak at sites within ≈6 km of the ILFZ. Shawmere B NRMs from this easterly zone tend to have higher inclinations and ≈1850 Ma VGP positions. *Symons and Vandall* [1990] sampled almost exclusively in this zone, and their average B inclinations are significantly steeper than ours.

TABLE 4. B_R and B_N Site Mean Directions for the Lackner Lake and Borden Carbonatites

Site	Declination, deg	Inclination, deg	*N/N'*	α_{95}, deg	Fisher precision, *k*
53 (L)	96	-39	1/6		
54 (L)	120	-29	2/8		12
54 (L)	313	50	4/8	22	10
55 (B)	301	52	3/5	11	56
68 (B)	313	49	1/5		

N' is the total number of samples per site; *N* is the number with stable behavior.

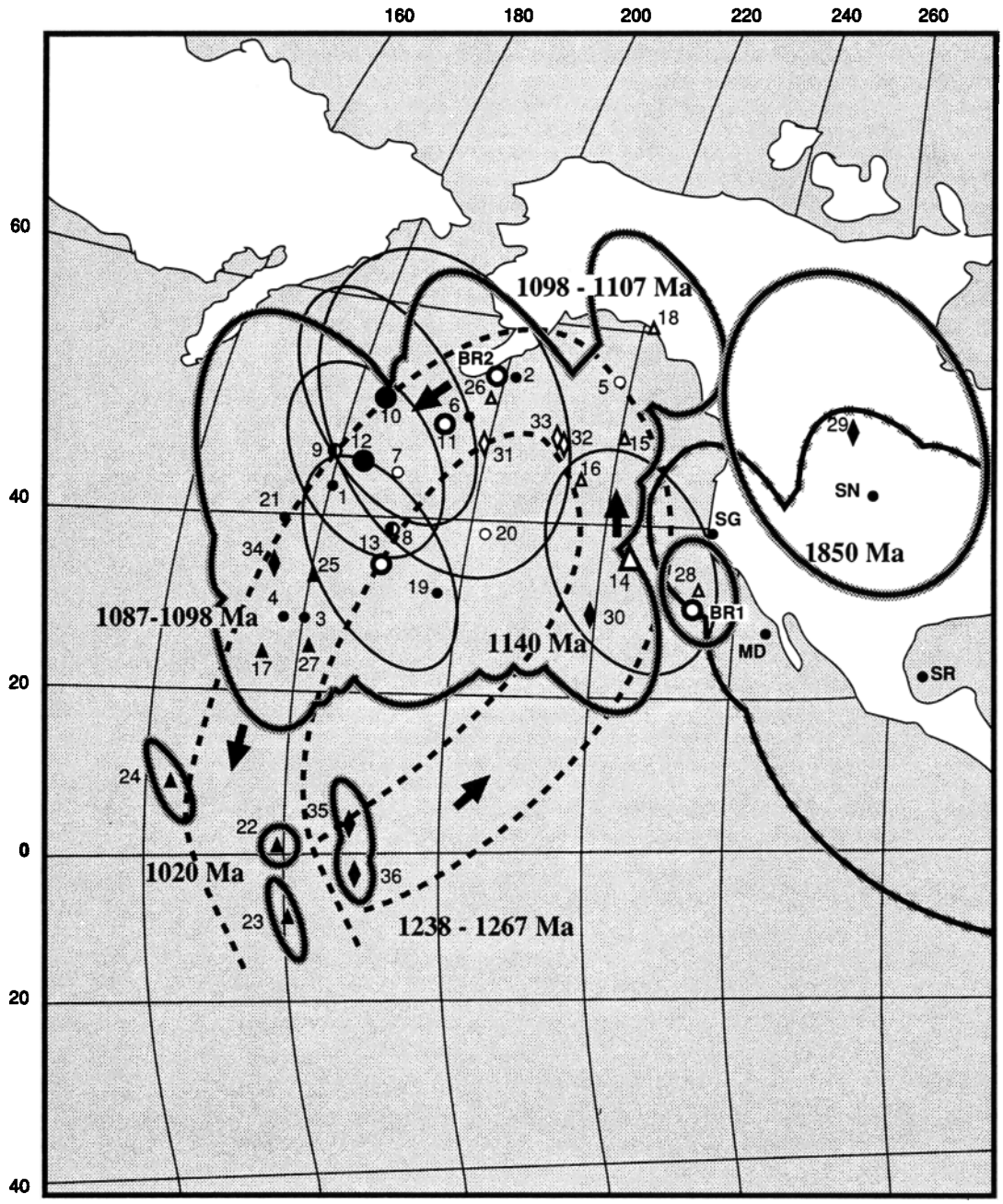


Fig. 6. Part of the middle Proterozoic apparent polar wander path for North America showing the paleopoles and 95% confidence ovals from this study (large triangle and circles), corresponding to B_N and B_R natural remanent magnetizations of the Shawmere anorthosite (14, B_{R1}, B_{R2}), the Shenango alkaline complex (12 and 13), and the Nemegosenda carbonatite (10 and 11). The updated version of the Logan loop has been constructed using the most reliable paleopoles for dikes (triangles) and other geological units in the Superior Province (diamonds). Also included are published paleopoles for Nemegosenda, Chipman, Seabrook, Lackner Lake, and Firesand River carbonatites and Clay-Howells and Coldwell alkaline complexes (circles) plus four paleopoles for the Superior Province with ages around 1850 Ma (MD, SG, SN, and SR). All these paleopoles are referenced in Table 5. Open and solid symbols correspond to reversed and normal polarities, respectively.

TABLE 5. Paleopoles for Keweenawan-Age (≈1108-1045 Ma) Alkaline Complexes, Dike Swarms, and Other Units and Hudsonian-Age (≈1850 Ma) Paleopoles From the Superior Province

	Unit	Polarity	Latitude °N	Longitude °E	*dp*	*dm*	Unit Age, Ma	Reference
				Keweenawan-Age Carbonatite and Alkaline Complexes				
1	Nemegosenda	N	43	182	15	24	1107 (U-Pb)	*Symons and Garber* [974]
2	Lackner Lake	N	54	203	7	8	1105 (U-Pb)	*Symons* [989*a*]
3	Firesand River	N	27	183	8	13	1097-1018 (K-Ar)	*Symons* [1989*b*]
4	Clay-Howells	N	27	179	4	8	1075 (Rb-Sr)	*Lewchuk and Symons* [1990*a*]
5	Coldwell 1	R	54	217	5	6	1108 (U-Pb)	*Lewchuk and Symons* [1990*b*]
6	Coldwell 2	N	49	195	7	9		*Lewchuk and Symons* [1990*b*]
7	Coldwell 3	R	44	189	5	7		*Lewchuk and Symons* [1990*b*]
8	Chipman Lake	N	38	186	7	11		*Symons* [1992]
9	Seabrook Lake	N	46	180	11	17		*Symons* [1992]
10	Nemegosenda	N	52	185	13	18	1107 (U-Pb)	this study
11	Nemegosenda	R	49	194	21	29		this study
12	Shenango	N	45	186	9	14	1045 (Rb-Sr)	this study
13	Shenango	R	34	191	11	17		this study
				Keweenawan-Age Intrusive, Sedimentary, and Volcanic Units				
14	Shawmere	R	38	226	13	16		this study
B_{R2}	Shawmere western sites (VGPs 1-4, Figure 7)	R	53	189	15	20		this study
B_{R1}	Shawmere eastern sites (VGPs 5-11, Figure 7)	R	30	235	15	17		this study
15	Logan sills	R	49	22	4	4.4	1109 (U-Pb)	*Halls and Pesonen* [1982]
16	Powder Mill Group	R	46	214	9	11		*Palmer and Halls* [1986]
17	Michipicoten Island	N	26	175	5	9	<1087 (U-Pb)	*Palmer and Davis* [1987]
18	Mamainse Point 4	R	61	219	13	14	>1087 (U-Pb)	*Palmer* [1970]
19	Mamainse Point 2	N	31	188	6	10		*Palmer* [1970]
20	Mamainse Point 3	R	37	203	15	20		*Palmer* [1970]
21	Mamainse Point 1	N	39	177	4	6		*Palmer* [1970]
22	Freda sandstone	N	1	180	1.4	3		*Henry et al.* [1977]
23	Jacobsville sandstone	N	−9	183	3	6		*Roy and Robertson* [1978]
24	Nonesuch shale	N	10	177	3	6		*Henry et al.* [1977]
25	Upper Osler volcanics	N	34	178	6	10	1108-1098 (U-Pb)	*Halls* [1974]
26	Lower Osler sequence	R	49	203	12	14		*Halls and Pesonen* [1982]
27	Portage Lake lavas	N	27	181	2	4	1096-1094 (U-Pb)	*Halls and Pesonen* [1982]
28	KSZ granulites	R	30	236	7	8		*Symons and Vandall* [1990]

					Keweenawan-Age Dike Swarms			
29	Groundhog River block	N	53	257	23	23.4		*Bates and Halls* [1991]
30	Abitibi dikes	N	32	220	9	11	1140.6 (U-Pb)	*Buchan and Halls* [1990]
31	Grand Portage	R	48	200	8	10		*Green et al.* [1987]
32	Marquette	R	48	213	5	6		*Pesonen and Halls* [1979]
33	Thunder Bay	R	48	212	7	8		*Pesonen* [1979]
34	Thunder Bay	N	35	174	4	7		*Pesonen* [1979]
35	Mackenzie	N	4	190	3	6	1267 (U-Pb)	*Buchan and Halls* [1990]
36	Sudbury	N	-3	192	2	4	1238.5 (U-Pb)	*Palmer et al.* [1977]
					Hudsonian-Age Paleopoles			
MD-Molson dikes		N	28	246	12	13	1883 (U-Pb)	*Ermanovics and Fahrig* [1975]
SR-Spanish River		N	37	264	24	25		*Robertson and Watkinson* [1974]
SN-Sudbury Norite		N	44	257	11	12	1850 (U-Pb)	*Morris* [1984]
SG-Sudbury granodiorite		N	40	234	9	10.5	1850.5 (U-Pb)	*Morris* [1984]

N is normal and *R* is reverse polarity; *dp* and *dm* are semi-minor and semi-major axes of the oval of 95% confidence.

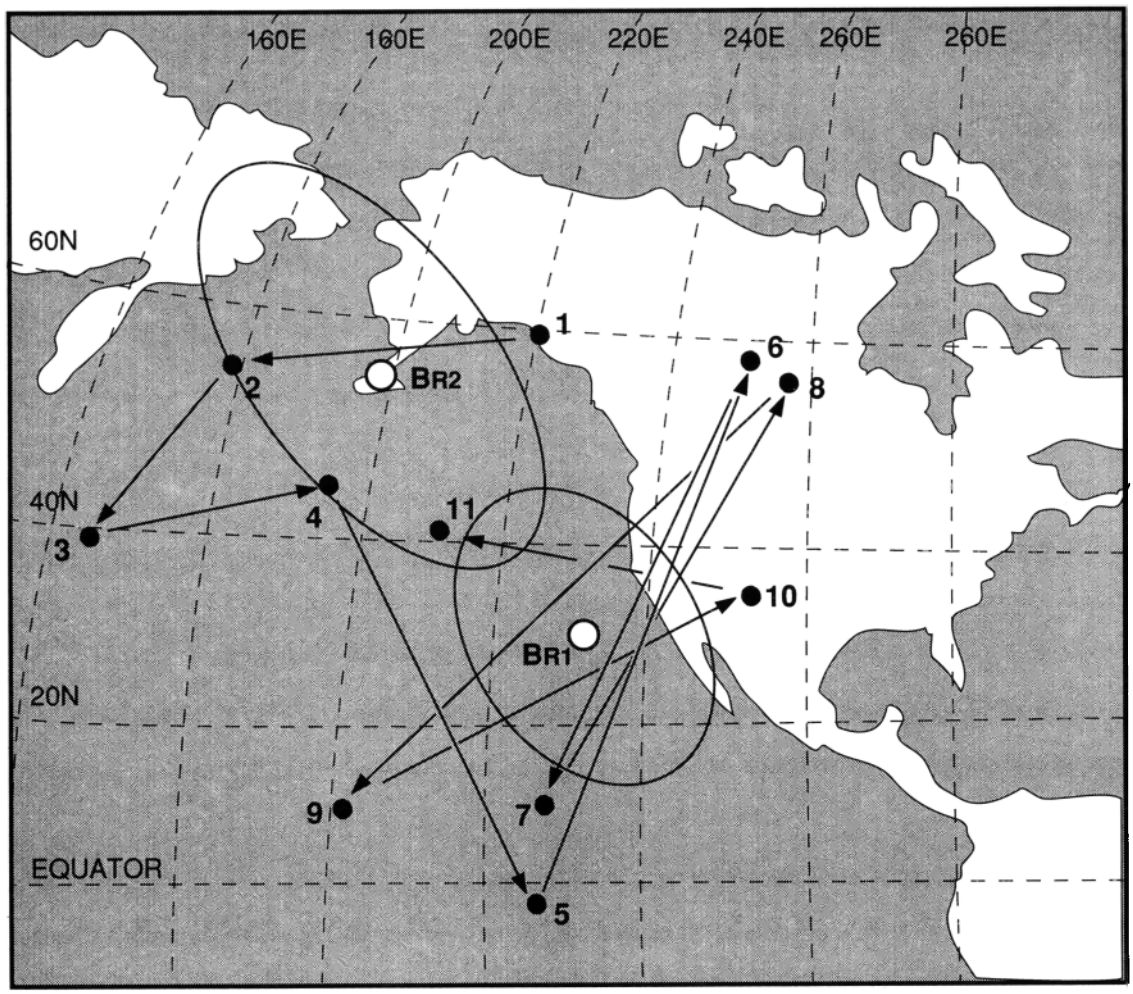


Fig. 7. Virtual geomagnetic poles (VGPs) for B_R sites in the Shawmere anorthosite plotted in order of their distance from the carbonatite line. The numbers correspond to sites in Figure 1 and Table 1 as follows: 1, SH4 and SH5; 2, 13; 3, 39; 4, 15; 5, SH3; 6, 40; 7, 17, SH2; 8, 19; 9, SH6; 10, LLL1-LLL5 and SH7; and 11, 42. Also included are average poles and 95% confidence ovals for the groupings of VGPs 1-4 (B_{R2}) and 5-11 (B_{R1}).

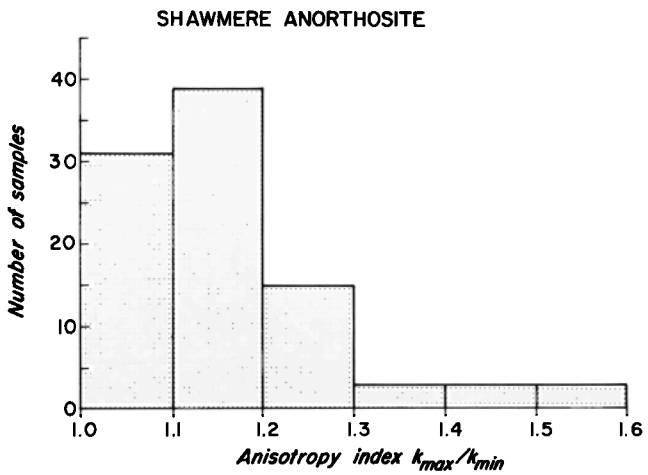


Fig. 8. Histogram of anisotropy indexes (k_{max}/k_{min}) in the Shawmere anorthosite showing a peak around 1.15.

8. Sites ≥6 km from either the ILFZ or the carbonatite line tend to be comparatively fresh. In some cases, they carry no B overprint at all. This central zone separates the two zones which seem to be characterized by middle Proterozoic and Keweenawan alteration, respectively.

These lines of evidence suggest to us that the Shawmere anorthosite bears two B overprints of rather different ages. The older remanence is more in evidence near the ILFZ, while to the west, Keweenawan-age hydrothermal alteration associated with intrusion of alkaline complexes has obliterated the early B to varying degrees and replaced it with a late B remanence of the same R polarity and similar direction. The two remanences are mixed in determining the average Shawmere B direction and may be mixed at the site and sample level also in some cases. Thus the average Shawmere B direction is probably not a single spot-reading of the geomagnetic field but a combination of two spot-readings at ≈1850 Ma and ≈1100 Ma, respectively.

On the basis of this interpretation, we can make the following reconstruction of the history of the Chapleau block. *Costanzo-Alvarez and Dunlop* [1988] and *Symons and Vandall* [1990] are in agreement that the A remanence of the gneisses and anorthosites was acquired ≈2550 Ma ago, prior to tilting. This remanence was preserved by the primary magnetic minerals.

Hanes et al. [1992] report a 2250-2300 Ma event, based on biotite and muscovite $^{40}Ar/^{39}Ar$ ages in both the footwall and the hanging wall adjacent to the ILFZ, which might be the age of main uplift, tilting, and thrusting of the Chapleau block. No paleomagnetic record of this event has survived.

Around 1880 Ma, in the earliest phase of incipient rifting and alkaline magmatism, the Borden and Cargill complexes were

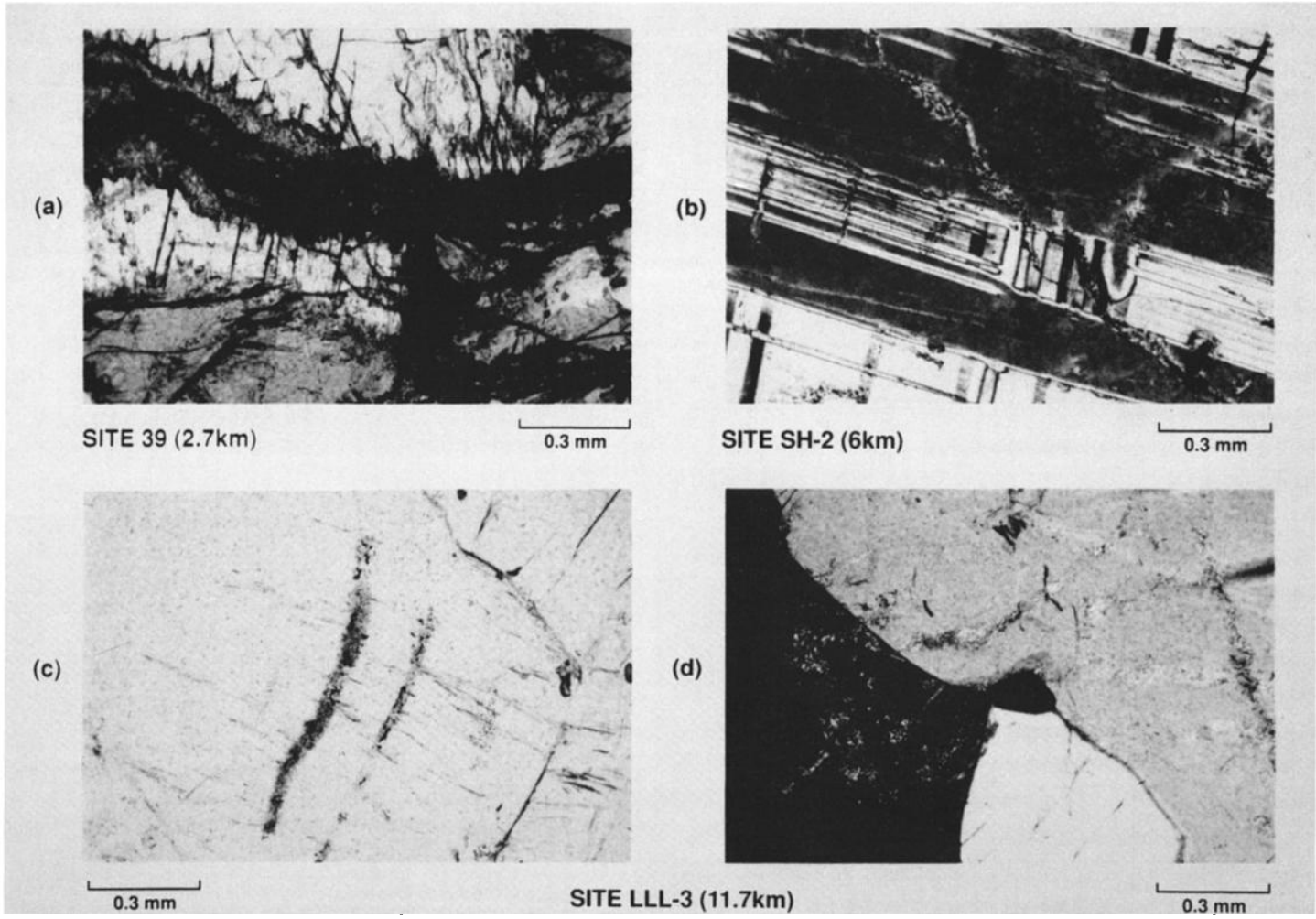


Fig. 9. Transmitted light photomicrographs for samples from sites 39, SH-2, and LLL-3 in the Shawmere anorthosite, located 2.7, 6, and 11.7 km, respectively, from the carbonatite line.

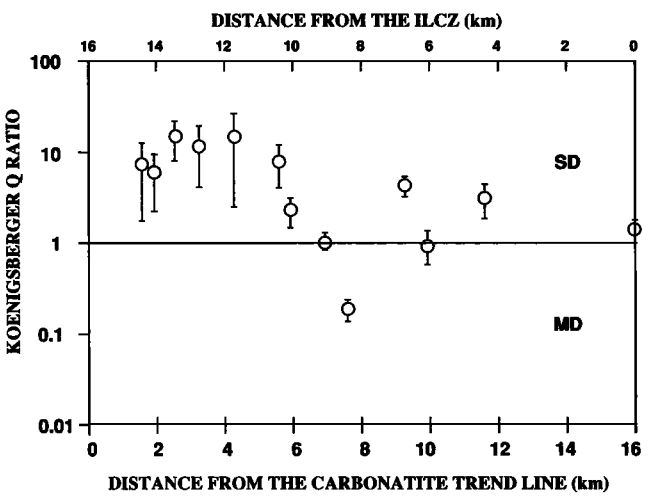


Fig. 10. Average Koenigsberger Q ratios (circles) and their ranges (vertical bars) for Shawmere anorthosite sites as a function of distance from major foci of hydrothermal alteration in the Chapleau block, the carbonatite line to the west, and the Ivanhoe Lake fault zone to the east. SD and MD indicate single-domain and multidomain characteristic ranges of Q values. The Q values and ranges include all five sites from Little Lemoine Lake and site SH-7 for a distance of 11.7 km from the carbonatite line, sites 17 and SH-2 for a distance of 6 km, and sites SH-4 and SH-5 for a distance of 1.7 km (Table 1).

intruded. Then or shortly after, biotite and muscovite $^{40}Ar/^{39}Ar$ ages in the Chapleau block were updated [*Hanes et al.*, 1992]. If this was also the time of uplift, as *Symons and Vandall* [1990] suggest, the uplift must have predated alkaline intrusion by at least 10-20 Ma, to allow time for the tectonic regime to change from major compression sufficient to transport KSZ thrust sheets 55-70 km to the east [*Geis et al.*, 1990; *Percival et al.*, 1989] to one of mild tension over a substantial fraction of the strike length of the KSZ. This event or events resulted in widespread overprinting of the A NRM in the Shawmere anorthosite by the early B_R remanence.

The final recorded event in the Chapleau block was intrusion of the second phase of alkaline bodies, whose primary TRMs we measure as the B NRMs of the Shenango, Nemegosenda, and other ≈1100 Ma KSZ carbonatites. As a result of renewed hydrothermal circulation, localized by fault systems along the western edge of the Shawmere body marking the carbonatite "rift" at depth, the westernmost sites in the Shawmere were completely overprinted by a younger B_R CRM/TCRM, while more distant sites were partially or not at all remagnetized.

CONCLUSIONS

1. The Shenango and Nemegosenda alkaline complexes carry primary TRM dating from ≈1100 Ma ago. Paleomagnetic poles are 45°N, 186°E (B_N) and 34°N, 191°E (B_R) for Shenango and 52°N, 185°E (B_N) and 49°N, 194°E (B_R) for Nemegosenda.

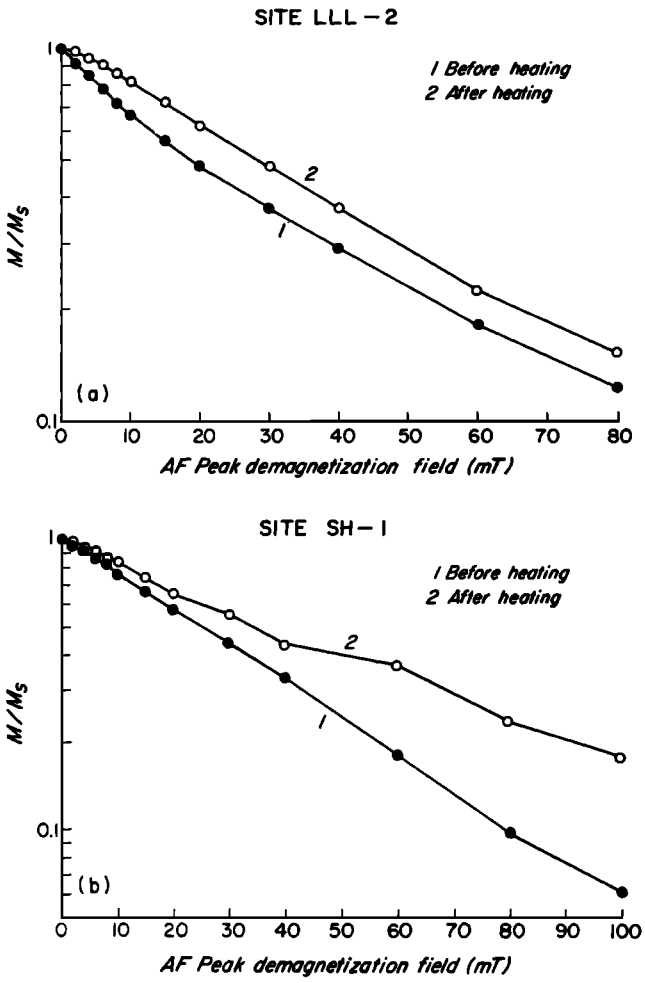


Fig. 11. Saturation isothermal remanence alternating field demagnetization curves for samples from sites LLL-2 and SH-1 before and after heating to 700°C.

2. Reverse-polarity remanences have been found for the Shenango, Nemegosenda, and Lackner Lake complexes. The R and N polarity NRMs are almost 180° reversed to each other, in contrast to Keweenawan paleopoles, which have a reversal asymmetry of ≈30°. This result throws into doubt the previously accepted predominance of asymmetric reversals of the Earth's field in Keweenawan time. It is possible that temporary quiescence in volcanic and magmatic activity in the Lake Superior area was responsible for the symmetric reversals not being found there. Short-term symmetric reversals can occur during periods in which the oscillating nondipole/dipole ratio assumes low values, approaching the behavior of an axial geomagnetic field. If on the other hand the difference between Keweenawan N and R poles is actually due to rapid APW, then the Shenango and Nemegosenda poles are tightly bracketed in age between about 1087 and 1107 Ma.

3. The Shawmere anorthosite B_R overprint has a paleopole at 38°N, 226°E that disagrees with the carbonatite poles, with Keweenawan R poles, and with middle Proterozoic cratonic poles.

4. Sites in the western part of the Shawmere body close to the "carbonatite line", a series of faults marking a reactivated zone of weakness, are severely altered. They have an abundance of fine-grained secondary magnetite and hematite, resulting in high Koenigsberger Q ratios. VGPs for these sites fall close to the Keweenawan track and define an average pole B_{R2} of 189°E, 53°N (dp=15°, dm=20°), suggesting that the B NRM is here a thermochemical overprint caused by hydrothermal fluids at the time of carbonatite intrusion.

5. Sites to the east are less altered. Their VGPs fall in the general area of ≈1850 Ma cratonic poles averaging to 235°E, 30°N (dp=15°, dm=17°) (B_{R1}). These B NRMs are probably TCRMs impressed by middle Proterozoic hydrothermal activity associated with intrusion of early alkaline bodies, fault movement on the ILFZ [*Ilanes et al.*, 1992], or late-stage uplift of the KSZ.

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